Community Digital Inclusion Survey

Data Cleaning & Analysis Report

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R Programming Capstone Project

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1. Executive Summary

This report presents the cleaning, analysis, and key findings from the Community Digital Inclusion Survey, covering 1,200 valid respondent records across four districts and six sectors. The raw dataset required standardising inconsistent text categories, correcting invalid numeric values, and removing duplicate records before analysis could proceed. The cleaned data was then used to answer six research questions on demographics, digital access, employment, and income, and to evaluate three participation and access targets set by the research team.

Overall, the survey met its female representation target but fell short on youth participation and, most notably, on internet access — driven by sharp disparities between districts and sectors rather than by gender. Education level emerged as the strongest driver of both income and digital skill levels. The relationship between training attendance and employment outcomes was inconclusive from this dataset alone and is discussed in Section 4.

2. Methodology & Data Cleaning

The raw dataset (1,214 records, 16 variables) was loaded and inspected in R using the tidyverse suite. Structure, missing values, and duplicate checks were run on every column before any changes were made. The table below summarises the major categories of cleaning applied; individual value-level corrections are not listed separately.

Issue Identified	Action Taken	Reason
14 duplicate respondent_id records	Removed duplicates, keeping the first occurrence	Each respondent should appear once; duplicates would double-count individuals in all summaries
Inconsistent text case/spacing in gender, sector, education_level, employment_status, and Yes/No fields (e.g. "male", "MALE", "Sector-A", "sector a")	Trimmed whitespace and standardised to a single consistent case/category per field	Inconsistent categories would be treated as separate groups, understating true category counts
Invalid ages (below 18 or above 100)	Recoded to missing (NA)	Values outside a plausible human age range indicate data entry errors

Issue Identified	Action Taken	Reason
Negative monthly_income values	Recoded to missing (NA)	Income cannot be negative; likely entry errors
satisfaction_score outside the valid 1–5 scale	Recoded to missing (NA)	Scores outside the documented scale range are invalid
Blank strings in education_level and digital_skill_level	Converted to explicit NA	Blank text is functionally missing data and should be treated consistently with other NAs

After cleaning, a derived variable, age_band, was created to group respondents into two categories (18–35 and 36+) to support the youth participation target evaluation in Section 4.

3. Research Questions & Findings

3.1 Demographic Profile of Respondents

The sample is 53.9% female and 46.1% male, with a mean age of 36 years (range 18–99). Respondents are concentrated in Gasabo (38.7%) and Nyagatare (26.5%) districts, with Kicukiro and Musanze making up the remainder. Secondary education is the most common attainment level (28.3%), followed by primary (24.6%) and university/tertiary (22.0%).

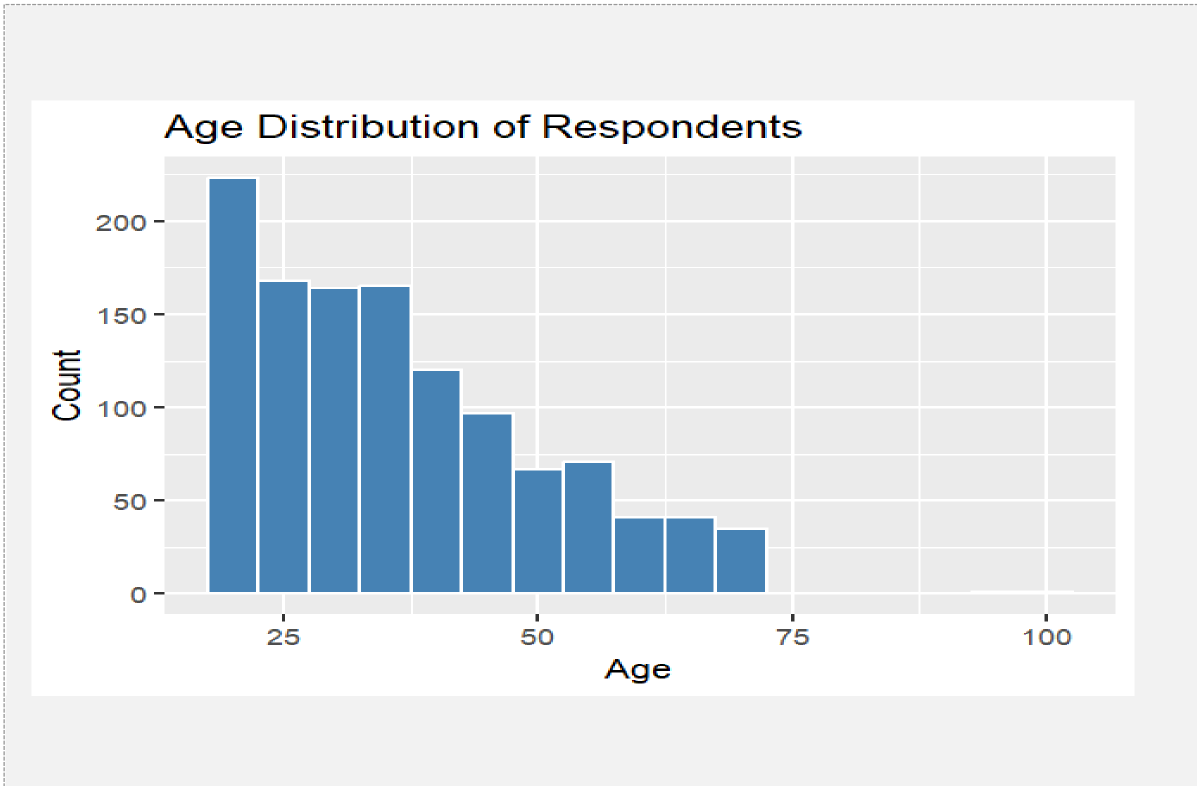

Figure 1. Age Distribution of Respondents — shows the spread of respondent ages and confirms the sample skews toward younger adults.

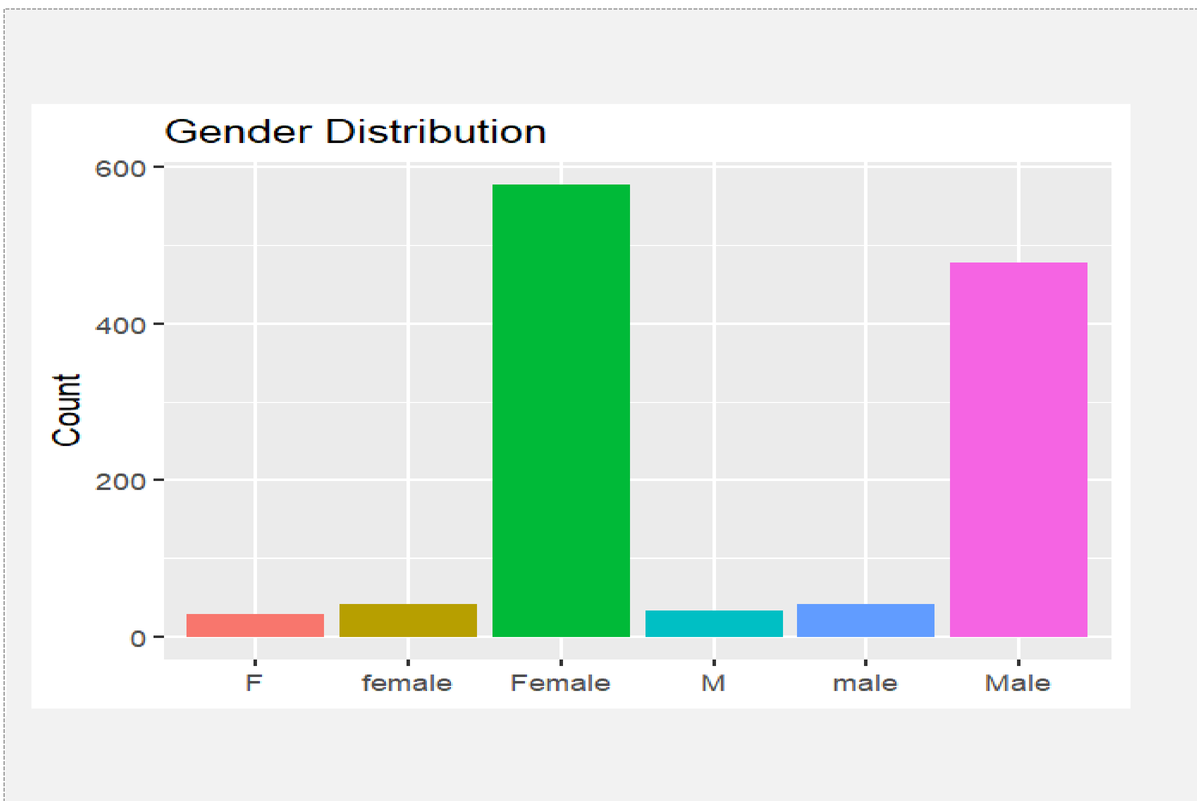

Figure 2. Gender Distribution — illustrates the near-even gender split described above.

3.2. Employment Status Overall and by Education Level

Overall, 41.9% of respondents are employed, 27.8% self-employed, 15.6% unemployed, and 14.8% students. This distribution shifts markedly with education: 59.8% of university-educated respondents are employed, compared with 24.1% of those with no formal education, who are more often self-employed (43.4%) or unemployed (28.9%).

3.3. Average Monthly Income by Education Level

Mean monthly income rises consistently with education level, from 62,480 RWF for respondents with no formal education to 256,722 RWF for those with university/tertiary education — more than a fourfold difference.

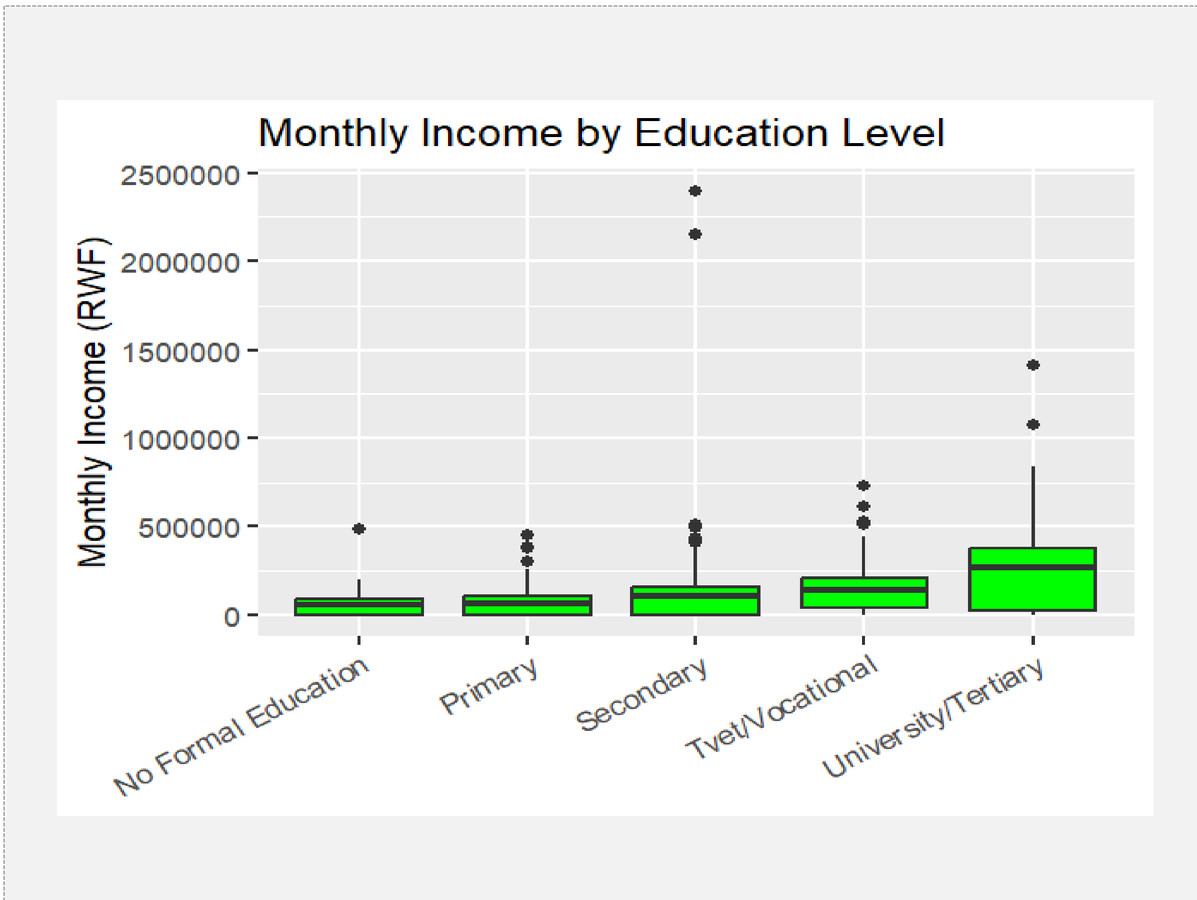

Figure 3. Monthly Income by Education Level — shows the income distribution for each education level and the scale of the income gap.

3.4.Training Attendance and Employment Outcomes

A direct comparison of current employment status shows little difference between respondents who attended training (39.0% currently employed or self-employed) and those who did not (43.5%). However, among respondents who attended training, 74.1% self-report that the training led to employment. This apparent contradiction most likely reflects that training was targeted at respondents who started from a weaker employment position (e.g. unemployed or student respondents), rather than the training having no effect.

Figure 4. Employment Rate by Training Attendance — compares the current employment rate of respondents who attended training against those who did not.

Interpretation note: because this is an observational survey rather than a controlled experiment, and respondents self-selected into the training programme, this dataset alone cannot establish that the training caused any observed difference in employment outcomes. Confounding factors — such as respondents' employment status prior to training, motivation, or access to opportunities — cannot be ruled out.

3.5. Groups with Lower Digital Access

Digital access gaps are driven far more by geography and education than by gender or age. Smartphone ownership, internet access, and advanced digital skills are all substantially lower in Nyagatare (43.4% / 38.1% / 2.8% respectively) than in Gasabo (81.5% / 80.6% / 28.4%). Similarly, advanced digital skills are reported by only 1.2% of respondents with no formal education, versus 33.0% of university-educated respondents. Gender differences across all three measures are minimal (within 2–3 percentage points).

4. Research Targets Evaluation

Target	Threshold	Result	Met?
Target 1 — Female Representation	≥ 50%	53.9%	Yes
Target 2 — Internet Access	≥ 70%	65.1%	No
Target 3 — Youth Participation (18–35)	≥ 60%	56.3%	No

Only the female representation target was met. Internet access fell short by nearly 5 percentage points overall, and is well below target in several sectors specifically (Section 3.2). Youth participation was close to, but under, its threshold.

5. Key Findings

- Digital access is primarily a geography problem, not a gender problem — internet access differs by up to 47.6 points between sectors, while the gender gap is under 1 point.
- Education level is the strongest driver of both income and job quality — university-educated respondents earn over four times more, on average, than those with no formal education.
- The training–employment relationship is ambiguous in this dataset: a raw current-employment comparison shows no advantage for trainees, but 74.1% of trainees self-report a positive outcome, suggesting a targeting effect rather than a training-failure effect.
- Advanced digital skills are concentrated among the young, educated, and urban — 33.0% among university-educated respondents versus 1.2% among those with no formal education, and 28.4% in Gasabo versus 2.8% in Nyagatare.
- Only one of three research targets was achieved (female representation); internet access and youth participation both fell short, with internet access showing the largest shortfall.

6. Recommendations

- Target infrastructure and connectivity investment geographically. Direct resources toward the lowest-access districts and sectors (notably Nyagatare, Musanze, and Sectors E/F) rather than gender-based programming, since access gaps track location far more than gender.
- Pair digital-skills training with foundational education support. Because income and advanced digital skills both rise sharply with education level,

bundling basic literacy/numeracy support with digital training may produce better outcomes for the least-educated respondents than digital training alone.

- Redesign training impact measurement before scaling the programme. Collect a pre-training employment baseline (or use a matched comparison group) so future evaluations can distinguish the training's actual effect from the effect of who was selected to attend it.

7. Closing Note

Of the issues identified, two stand out as priorities for the next phase. First, the scale of the geographic access gap — nearly 48 points between the best- and worst-served sectors — means that broad, evenly distributed interventions are unlikely to close it; targeting is essential. Second, the current survey design cannot support a causal claim about the training programme's effectiveness, which limits confidence in reported outcomes; a small design change (adding a pre-training baseline) would substantially strengthen the evidence available for the next evaluation.